

# Soham Narsale

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## EDUCATION

### Pimpri Chinchwad College of Engineering

Bachelor of Technology in Computer Science and Engineering (AI & ML)

- Grade: 8.33 CGPA

Pune, Maharashtra

December 2021 - Present

## EXPERIENCE

### AI/ML Intern

CaretCloud

July 2024 – October 2024

Bangalore, India

- Developed a Retrieval-Augmented Generation (RAG) chatbot for compliance, risk management and governance using MistralAI, FAISS, LangChain, and FastAPI, and deployed the solution on AWS EC2, ensuring scalability.
- Engineered a Generative AI-powered document generation system using OpenAI's GPT-4o, LlamaIndex, and AWS Lambda, to automate compliance document creation based on structured database inputs.
- Implemented a multi-tenant RAG system for tenant-specific policy document querying. Utilized OpenAI's GPT-4o, LlamaIndex, AWS Lambda, and AWS S3 ensuring secure data segregation for seamless multitenancy.

### AI/ML Intern

Muks Robotics

April 2024 – July 2024

Pune, India

- Developed and trained advanced object detection and computer vision models for face recognition and anomaly detection projects, resulting in a 25% increase in detection accuracy.
- Engineered and implemented a voice assistance chatbot using state-of-the-art pre-trained models such as LLaMA 2, OpenAI Whisper, and pyttsx3, enhancing user interaction and accessibility.
- Created an innovative image-to-text chatbot utilizing LLaVA and MoonDream 2, facilitating seamless speech-to-speech communication and improving user experience by 40%.

## PROJECTS

### Generative AI-powered Internal Knowledge Sharing Platform

- Designed platform facilitating employee and SME collaboration through Q&A, tagging, categorization, and up-vote/downvote features for quality content curation.
- Implemented Role-Based Access Control (RBAC) for secure logins with distinct roles for SMEs and Employees.
- Built a RAG system to generate AI-powered contextual responses using LlamaIndex and Groq AI Inference Engine with the Llama-3.1-8B-instant model.
- Utilized HuggingFace Embeddings for semantic search and content categorization, enhancing the knowledge discovery process.

### Multimodal Conversational AI Assistant

- Developed a voice assistant with multimodal LLM Llava 1.5 7B, enhancing image and text understanding.
- Integrated OpenAI's Whisper for precise speech-to-text conversion, improving interaction and accessibility.
- Implemented realistic text-to-speech functionality with gTTS, elevating user experience.
- Deployed a user-friendly Gradio app, ensuring seamless functionality and interface integration.

## TECHNICAL SKILLS

**Technologies:** Generative AI, Natural Language Processing, Computer Vision, Deep Learning, Machine Learning

**Libraries:** Numpy, Pandas, Matplotlib, Scikit-learn, TensorFlow, Transformers, NLTK, Spacy, OpenCV, YOLO

**Languages and Frameworks:** Python, Langchain, LlamaIndex, OpenAI, Streamlit, Flask, FastAPI

**Developer Tools and Version Control:** VS Code, Git, GitHub, AWS, Google Cloud, Postman, Jira, Confluence

## CERTIFICATIONS

Generative AI with Large Language Models - AWS

June 2024

Advanced Computer Vision with TensorFlow - DeepLearning.AI

January 2024

Convolutional Neural Networks - DeepLearning.AI

December 2023

AWS Cloud Practitioner - AWS

May 2024

Machine Learning A-Z: AI, Python & R - Udemy

December 2023

Getting Started with AI on Jetson Nano - NVIDIA

July 2023